

Retail Sales Analytics Project – Key Insights & Submission Notes

Dashboard Components (Power BI)

1. Region-wise Sales Breakdown (Pie/Bar Chart)
2. Monthly Sales Trend (Line Chart)
3. Top 5 Products (Bar Chart)
4. Category Performance Trend
5. Sales Channel Distribution
6. KPI Cards: Total Sales, Total Transactions, Unique Customers, Average Order Value
7. Slicers: Region, Month, Sales Channel

Project KPIs

- Total Sales: 111.58 Million
- Total Transactions: 5,000
- Unique Customers: 2,419
- Average Order Value (AOV): 22,315.79

5 Key Insights

1. Electronics generated the highest revenue among all categories, contributing the largest share of sales.
2. Tablet, Shoes, Notebook, TV, and Fan emerged as the top-performing products by revenue.
3. The dataset contains 5,000 transactions from 2,419 unique customers, indicating a strong repeat-customer base.
4. Average Order Value exceeded 22K, showing relatively high-value transactions.
5. Regional, channel, and category filters in Power BI help identify profitable segments and seasonal trends.

SQL Queries Used

Total Sales: `SELECT SUM(TotalAmount) AS TotalSales FROM RetailTransaction;` Total Transactions: `SELECT COUNT(*) AS TotalTransactions FROM RetailTransaction;` Unique Customers: `SELECT COUNT(DISTINCT CustomerID) AS UniqueCustomers FROM RetailTransaction;` Average Order Value: `SELECT SUM(TotalAmount)/COUNT(*) AS AOV FROM RetailTransaction;` Top 5 Products: `SELECT ProductName, SUM(TotalAmount) AS Sales FROM RetailTransaction GROUP BY ProductName ORDER BY Sales DESC LIMIT 5;`

Power BI Implementation

- Imported CSV into Power BI.
- Created relationships and date hierarchy.
- Built KPI cards for Sales, Transactions, Customers, and AOV.
- Used Line, Bar, and Pie visuals for trend and distribution analysis.
- Added Region, Month, and Sales Channel slicers for interactive reporting.